

Problem 25.3

If

$$\bar{a}_{\overline{n}|} = 4 \quad \text{and} \quad \bar{s}_{\overline{n}|} = 12,$$

find δ .

Solution

For a continuous annuity, the accumulated value and present value are related by

$$\bar{s}_{\overline{n}|} = \bar{a}_{\overline{n}|} e^{\delta n}.$$

Given

$$\bar{a}_{\overline{n}|} = 4 \quad \text{and} \quad \bar{s}_{\overline{n}|} = 12,$$

we have

$$12 = 4e^{\delta n}.$$

Therefore,

$$e^{\delta n} = 3.$$

Also,

$$\bar{s}_{\overline{n}|} = \frac{e^{\delta n} - 1}{\delta}.$$

Substitute $\bar{s}_{\overline{n}|} = 12$ and $e^{\delta n} = 3$:

$$12 = \frac{3 - 1}{\delta}.$$

Thus,

$$12 = \frac{2}{\delta}.$$

Solving for δ ,

$$\delta = \frac{2}{12} = \frac{1}{6}.$$

Therefore,

$$\boxed{\delta = \frac{1}{6}}.$$